

STUDENT UTILITY PROGRAM

Java Programming Practical Record

Question	Implementation of basic Java programming constructs using a menu-driven Student Utility Program.
Aim	To understand and implement variables, data types, operators, conditional statements, loops, input/output, and parameterized user-defined methods in Java.
Submitted by	G. Pranadeep
Roll No.	A25126552212
Class	2/4 CSM-D
Branch	Computer Science (Artificial Intelligence / Machine Learning)

Concepts Covered

- Java Program Structure
- Variables and Data Types
- Operators
- Input and Output
- Conditional Statements: if, if-else, switch
- Looping Statements: for, while, do-while
- Methods (User-defined Functions)

Task: Develop a menu-driven Student Utility Program that performs the operations specified in Parts A–D of the question.

1. Problem Statement

Develop a **Student Utility Program** using a menu-driven approach. The program must accept student details, perform conditional and looping operations, and call parameterized methods that return results.

Part A: Java Basics

Display student information by accepting Name, Roll Number, and Marks in three subjects. Calculate Total and Percentage and display the result.

Part B: Conditional Statements

1. Check whether a number is Even or Odd.
2. Find the Largest of Three Numbers.
3. Display Grade based on percentage using if-else.
4. Display Day of the Week using switch.

Part C: Looping Statements

1. Print multiplication tables.
2. Display numbers from 1 to N.
3. Calculate the sum of first N natural numbers.
4. Generate the Fibonacci series.

Part D: Methods

Create parameterized, returning methods for factorial, prime checking, maximum of two numbers, and area of a circle. Call each method from main().

2. Steps for Execution

1. Open a Java development environment such as VS Code, IntelliJ IDEA, Eclipse, or a command prompt with JDK installed.
2. Create a Java source file named StudentProgram.java.
3. Copy the complete program into the file and save it.
4. Compile the source file using `javac StudentProgram.java`.
5. Run the compiled program using `java StudentProgram`.
6. Choose an option from the displayed menu and enter the requested values.
7. Continue using the menu until option 14 (Exit) is selected.

3. Algorithm

1. Start the program and create a Scanner object for keyboard input.
2. Create a StudentProgram object so that the user-defined methods can be called.
3. Display the Student Utility Program menu and read the user's choice.
4. For student information, read name, roll number, and three marks; calculate total and percentage.
5. For conditional operations, use if/else or switch according to the selected option.
6. For looping operations, use for loops to generate tables, numbers, sums, and Fibonacci terms.
7. For method operations, pass the required arguments to factorial(), isPrime(), max(), and area(), then display the returned result.
8. Repeat the menu using do-while until the user chooses option 14.
9. Close the Scanner and stop the program.

4. Java Program Structure

The program follows the basic Java structure: import statement, class declaration, instance methods, the main() method, object creation, input using Scanner, a do-while menu loop, switch cases, and program termination.

5. Java Source Code

Filename: StudentProgram.java

```
import java.util.Scanner;

public class StudentProgram {
    int factorial(int n) {
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact *= i;
        }
        return fact;
    }

    boolean isPrime(int n) {
        if (n <= 1)
            return false;
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0)
                return false;
        }
        return true;
    }

    int max(int a, int b) {
        if (a > b)
            return a;
        else
            return b;
    }

    double area(double r) {
        return 3.14 * r * r;
    }

    public static void main(String[] args) {
        StudentProgram s = new StudentProgram();
        Scanner sc = new Scanner(System.in);
        int ch;

        do {
            System.out.println("\nSTUDENT UTILITY PROGRAM");
            System.out.println("1.Student Information");
            System.out.println("2.Even or Odd");
            System.out.println("3.Largest of Three Numbers");
            System.out.println("4.Display Grade");
            System.out.println("5.Day of the Week");
            System.out.println("6.Multiplication Table");
            System.out.println("7.Numbers from 1 to N");
            System.out.println("8.Sum of First N Natural Numbers");
            System.out.println("9.Fibonacci Series");
            System.out.println("10.Factorial");
            System.out.println("11.Prime Check");
            System.out.println("12.Maximum of Two Numbers");
            System.out.println("13.Area of Circle");
            System.out.println("14.Exit");
            System.out.println("Enter your choice :");
            ch = sc.nextInt();

            switch (ch) {
                case 1:
                    sc.nextLine();
                    System.out.println("Enter Student Name: ");
                    String name = sc.nextLine();
                    System.out.println("Enter Roll Number: ");
                    int roll = sc.nextInt();
                    System.out.println("Enter Marks in Subject 1: ");
                    double m1 = sc.nextDouble();
                    System.out.println("Enter Marks in Subject 2: ");
                    double m2 = sc.nextDouble();
                    System.out.println("Enter Marks in Subject 3: ");
                    double m3 = sc.nextDouble();
                    double total = m1 + m2 + m3;
                    double perc = total / 3;
                    System.out.println("\nStudent Details");
                    System.out.println("Name      : " + name);
                    System.out.println("Roll No   : " + roll);
                    System.out.println("Total Marks: " + total);
                    System.out.println("Percentage : " + perc + "%");
                    break;
            }
        } while (ch != 14);
    }
}
```

5. Java Source Code (continued)

```
case 2:
    System.out.println("Enter a Number: ");
    int num = sc.nextInt();
    if (num % 2 == 0)
        System.out.println("Even Number");
    else
        System.out.println("Odd Number");
    break;

case 3:
    System.out.println("Enter First Number: ");
    int a = sc.nextInt();
    System.out.println("Enter Second Number: ");
    int b = sc.nextInt();
    System.out.println("Enter Third Number: ");
    int c = sc.nextInt();
    int largest = a;
    if (b > largest)
        largest = b;
    if (c > largest)
        largest = c;
    System.out.println("Largest Number = " + largest);
    break;

case 4:
    System.out.println("Enter Percentage: ");
    double per = sc.nextDouble();
    if (per >= 90)
        System.out.println("Grade A");
    else if (per >= 75)
        System.out.println("Grade B");
    else if (per >= 60)
        System.out.println("Grade C");
    else if (per >= 40)
        System.out.println("Grade D");
    else
        System.out.println("Grade F");
    break;

case 5:
    System.out.println("Enter Day Number (1-7): ");
    int day = sc.nextInt();
    switch (day) {
        case 1:
            System.out.println("Monday");
            break;
        case 2:
            System.out.println("Tuesday");
            break;
        case 3:
            System.out.println("Wednesday");
            break;
        case 4:
            System.out.println("Thursday");
            break;
        case 5:
            System.out.println("Friday");
            break;
        case 6:
            System.out.println("Saturday");
            break;
        case 7:
            System.out.println("Sunday");
            break;
        default:
            System.out.println("Invalid Day");
    }
    break;

case 6:
    System.out.println("Enter Number: ");
    int numb = sc.nextInt();
    System.out.println("Multiplication Table:");
    for (int i = 1; i <= 10; i++) {
        System.out.println(numb + " x " + i + " = " + (numb * i));
    }
    break;
```

5. Java Source Code (continued)

```

        case 7:
            System.out.println("Enter N: ");
            int n = sc.nextInt();
            for (int i = 1; i <= n; i++) {
                System.out.println(i + " ");
            }
            System.out.println();
            break;

        case 8:
            System.out.println("Enter N: ");
            int numN = sc.nextInt();
            int sum = 0;
            for (int i = 1; i <= numN; i++) {
                sum += i;
            }
            System.out.println("Sum = " + sum);
            break;

        case 9:
            System.out.println("Enter Number of Terms: ");
            int terms = sc.nextInt();
            int first = 0, second = 1;
            System.out.println("Fibonacci Series:");
            for (int i = 1; i <= terms; i++) {
                System.out.println(first);
                int next = first + second;
                first = second;
                second = next;
            }
            System.out.println();
            break;

        case 10:
            System.out.println("Enter Number: ");
            int fact = sc.nextInt();
            System.out.println("Factorial = " + s.factorial(fact));
            break;

        case 11:
            System.out.println("Enter Number: ");
            int prime = sc.nextInt();
            if (s.isPrime(prime))
                System.out.println("Prime Number");
            else
                System.out.println("Not a Prime Number");
            break;

        case 12:
            System.out.println("Enter First Number: ");
            int x = sc.nextInt();
            System.out.print("Enter Second Number: ");
            int y = sc.nextInt();
            System.out.println("Maximum = " + s.max(x, y));
            break;

        case 13:
            System.out.println("Enter Radius: ");
            double radius = sc.nextDouble();
            System.out.println("Area = " + s.area(radius));
            break;

        case 14:
            System.out.println("Exiting Program...");
            break;

        default:
            System.out.println("Invalid Choice!");
    }
} while (ch != 14);

sc.close();
}

```

6. Methods Implemented

Method	Parameters	Return Type	Purpose
factorial(int n)	n	int	Returns factorial of n using a for loop.

isPrime(int n)	n	boolean	Checks whether n is prime.
max(int a, int b)	a, b	int	Returns the larger of two integers.
area(double r)	r	double	Returns the area of a circle using $3.14 \times r \times r$.

7. Representative Sample Output

Sample run demonstrating student information, conditional operations, looping operations, and methods:

```
STUDENT UTILITY PROGRAM
1.Student Information
2.Even or Odd
3.Largest of Three Numbers
4.Display Grade
5.Day of the Week
6.Multiplication Table
7.Numbers from 1 to N
8.Sum of First N Natural Numbers
9.Fibonacci Series
10.Factorial
11.Prime Check
12.Maximum of Two Numbers
13.Area of Circle
14.Exit
Enter your choice : 1

Enter Student Name:
Pranadeep
Enter Roll Number:
101
Enter Marks in Subject 1:
85
Enter Marks in Subject 2:
90
Enter Marks in Subject 3:
80

Student Details
Name      : Pranadeep
Roll No   : 101
Total Marks: 255.0
Percentage : 85.0%

Enter your choice : 10
Enter Number:
5
Factorial = 120

Enter your choice : 11
Enter Number:
17
Prime Number

Enter your choice : 12
Enter First Number:
25
Enter Second Number:
40
Maximum = 40

Enter your choice : 13
Enter Radius:
7
Area = 153.86

Enter your choice : 14
Exiting Program...
```

8. Observation Questions and Answers

1. What is the structure of a Java program?

Answer: A Java program generally contains import statements, a class declaration, variables and methods, and a main() method from which execution begins. Statements are written inside methods and grouped using braces.

2. Differentiate between primitive and non-primitive data types.

Answer: Primitive data types directly store simple values, such as int, double, char, and boolean. Non-primitive data types refer to objects or classes, such as String, arrays, and user-defined classes.

3. Compare if-else and switch statements.

Answer: if-else is suitable for conditions involving ranges, expressions, or multiple logical comparisons. switch is convenient when one expression is compared against several fixed case values.

4. Explain the differences between for, while, and do-while loops.

Answer: A for loop is commonly used when initialization, condition, and update are known together. A while loop checks its condition before each iteration and may execute zero times. A do-while loop checks the condition after the body, so it executes at least once.

5. What are methods? Why are they useful in Java?

Answer: Methods are named blocks of code that perform a specific task. They improve code reuse, organization, readability, and modularity. In this program, the four methods return results to main().

6. Explain the difference between formal parameters and actual parameters.

Answer: Formal parameters are the variables written in a method definition, such as n in factorial(int n). Actual parameters are the values or variables supplied when the method is called, such as factorial(fact).

9. Marks Distribution

Component	Marks
Java Program Structure	2
Conditional Statements	2
Looping Statements	2
Methods Implementation	3
Observation Answers	1
Total	10

10. Learning Outcomes

Students will be able to:

- Develop basic Java programs.
- Apply decision-making and looping constructs.
- Design reusable methods.
- Improve logical thinking and problem-solving skills.

Result

The menu-driven Student Utility Program was implemented successfully. The program demonstrates Java program structure, variables and data types, input/output, conditional statements, looping statements, and parameterized user-defined methods that return results.

Student Signature

Faculty Signature
